

Электрическое поле

① min. заряд $1,6 \cdot 10^{-19} \text{ Кл}$

Закон сохранения заряда

② Потенциальность э. поля

$$\vec{E} \quad \vec{F} = q\vec{E}$$

③ Закон Кулона

$$\text{СИ: } F = k \frac{q_1 q_2}{r^2} \quad k = 9 \cdot 10^9 \quad [q] = \text{Кл}$$

$$\text{СГСЭ: } F = \frac{q_1 q_2}{r^2} \quad [q] = \sqrt{\text{дин} \cdot \text{см}}$$

④ $E = \frac{q}{r^2} \text{ (СГСЭ)}$

$$\vec{E} = \frac{q}{r^2} \vec{e}_r \quad \vec{e}_r = \frac{\vec{r}}{r}$$

$$\vec{E} = \frac{q}{r^3} \vec{r}$$

⑤ Принцип суперпозиции

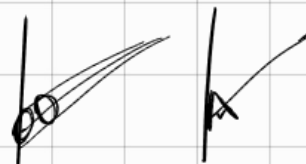
$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \dots$$

заряд / eq. поверх.

$$dE_{\perp} = \frac{dq}{r^2} \cos \theta = \frac{\sigma dS \cos \theta}{r^2}$$

поверх. перп. и пог. вект.

$$d\Omega = \frac{dS \cos \theta}{r^2} \quad \uparrow dS_{\perp}$$



$$E_{\perp} = \sigma \Omega$$

угловой: градус



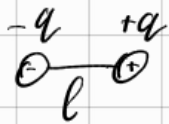
$$\Omega = 2\pi (1 - \cos \theta)$$

Электрический диполь



$$\vec{p} = \sum_{i=1}^N q_i \vec{r}_i$$

$$Q = \sum_{i=1}^N q_i = 0$$



- эл. диполь $\vec{p} = ql$

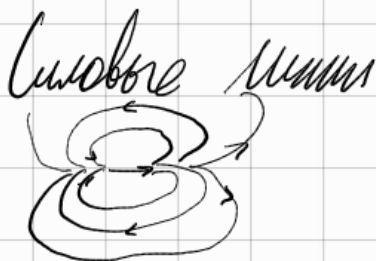
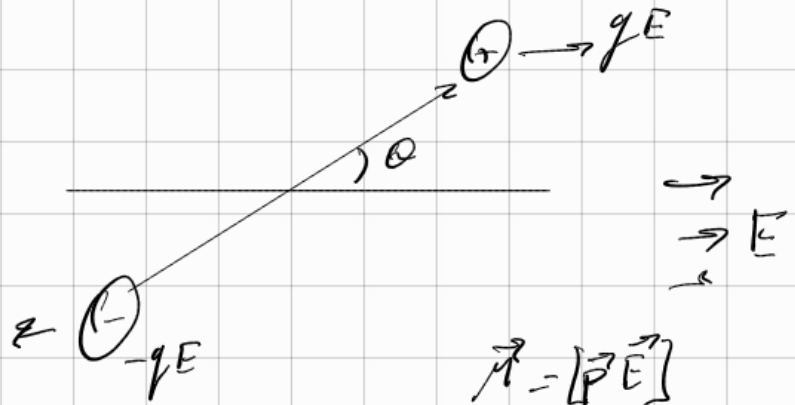
Эл. поле диполя



$$\vec{E}(\vec{r}) = \frac{3(\vec{p} \cdot \vec{r})\vec{r}}{r^5} - \frac{\vec{p}}{r^3}$$

Момент сил

$$M = pE \sin \theta$$



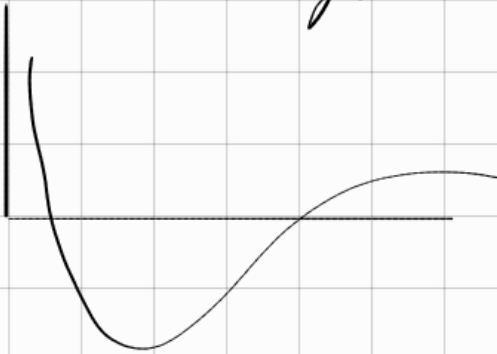
Сила в-з-я 2 зарядов

$$F = p \frac{dE}{dr}$$

$$\vec{F} = -q \vec{E}(r) + q \vec{E}(r+l)$$

$$F = P \frac{dE}{dr} \quad E = \frac{2P_1}{r^3} \quad F = -\frac{6P_1P_2}{r^4}$$

Потенциал Ланжана - Дирака



$$U = \frac{Q_1}{r^{1/2}} - \frac{Q_2}{r^6}$$

$$F = -\frac{dU}{dr} \sim \frac{1}{r^2}$$

Результат

Пояснение к

1) Угловое взаимодействие

$$P_1 \neq 0 \quad P_2 = 0$$

$$P_2 = \alpha E = \frac{2\alpha P_1}{r^3}$$

$$\begin{matrix} + & - \\ \leftarrow & \\ \leftarrow & \\ \leftarrow & \end{matrix} \quad P = \alpha E$$

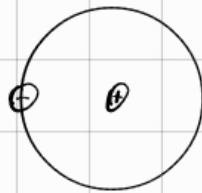
, если поперечность в $F = \frac{6P_1P_2}{r^4}$, то $\sim \frac{1}{r^2}$

2) Ориентационное взаимодействие

$$P_1 \neq 0 \quad P_2 \neq 0 \quad P_2 \langle \cos \theta \rangle \sim \frac{P_2}{r^3}$$

3) Дипольное взаимодействие

$$P_1 = 0 \quad P_2 = 0$$



Теорема Гаусса

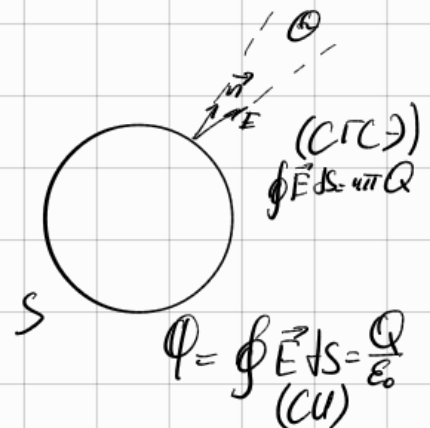
$$\vec{S} \rightarrow \vec{v}$$

$$\Phi = \vec{S} \cdot \vec{v}$$

$$\Phi = \vec{v} \cdot \vec{S} \cdot \cos \theta = \vec{v} \cdot \vec{S}$$

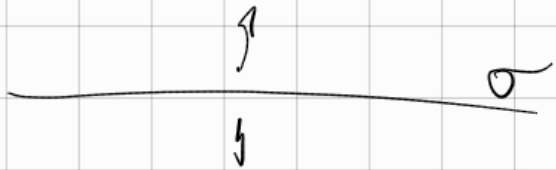
$$\vec{S} = \vec{S} \cdot \vec{v}$$

$$\Phi = \int_S \vec{E} \cdot \vec{S} = \int_S E_n dS$$

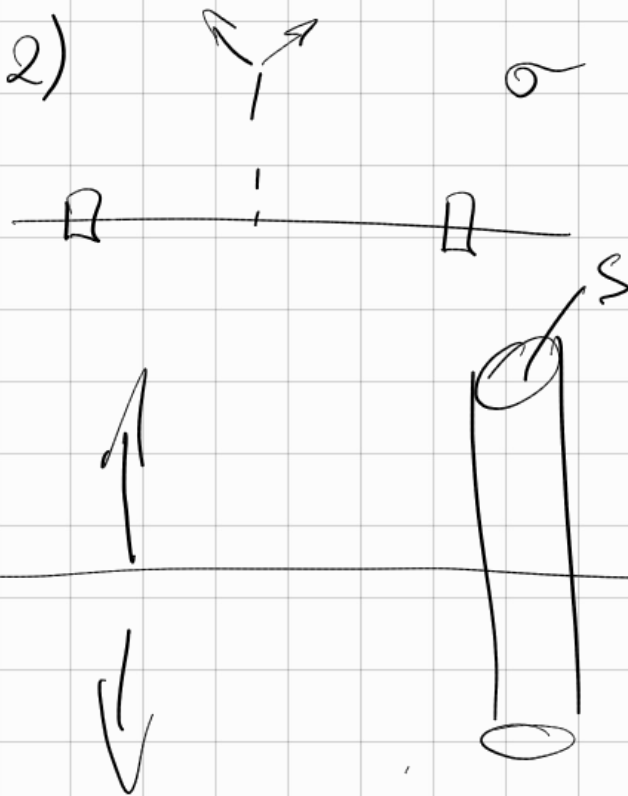
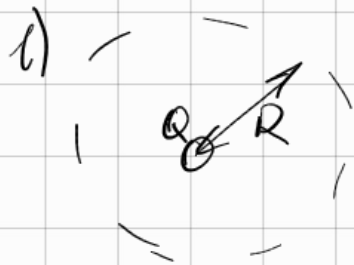



 $E = \frac{Q}{r^2}$
 $\Phi = E 4\pi r^2 \Rightarrow Q = 4\pi r^2 E$

Plane electrodes

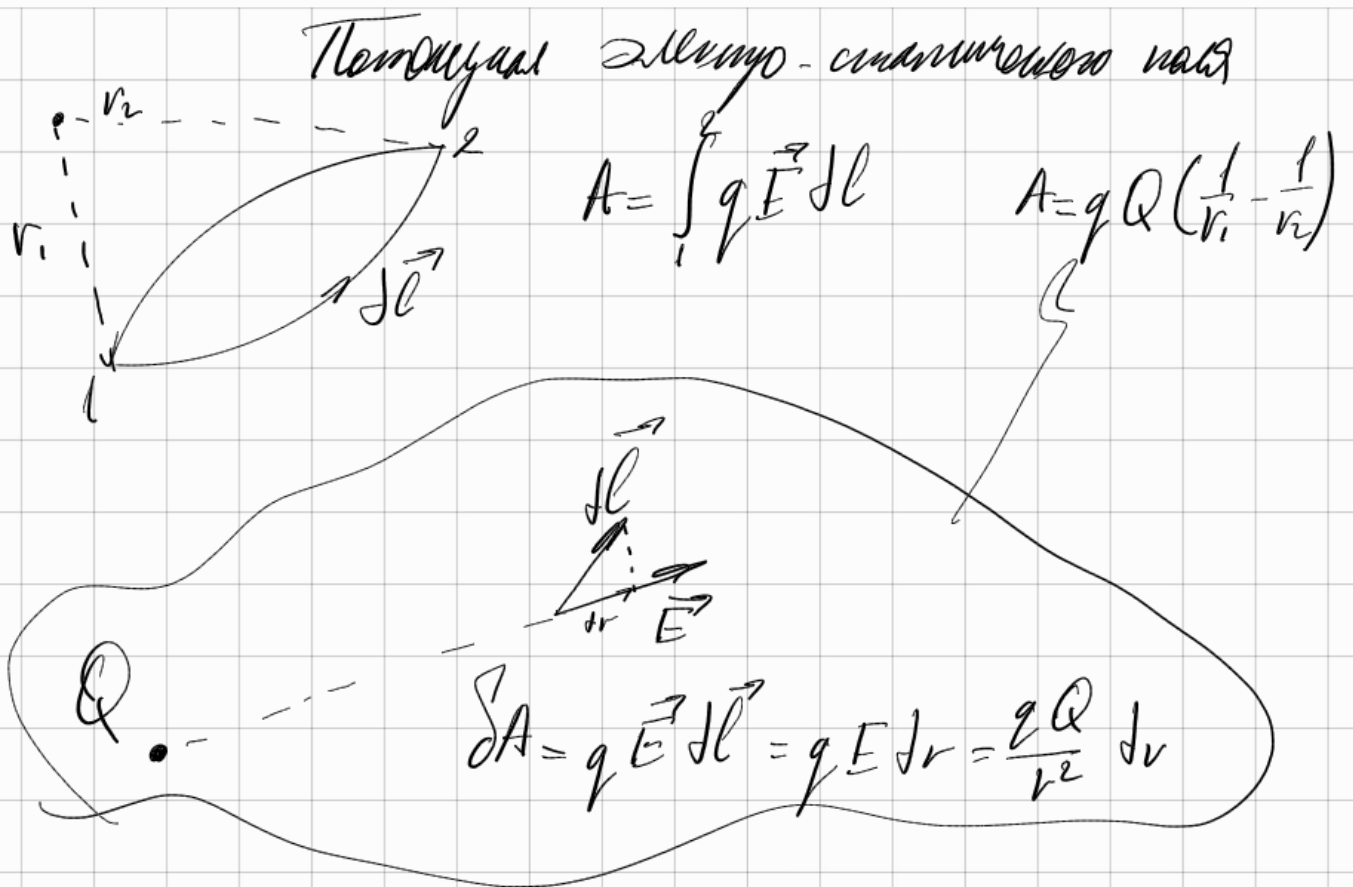

 $E = \sigma / 2\epsilon_0$

$\oint \vec{E} \cdot d\vec{S} = 4\pi Q \quad \text{CTC} \Rightarrow E \cdot 4\pi R^2 = 4\pi Q \Rightarrow E = \frac{Q}{R^2}$



$\Phi = 2ES = 4\pi\sigma S$
 $\Rightarrow E = 2\pi\sigma$


 $\Phi = 2\pi vl E = 4\pi vl$
 $\Rightarrow E = 2\lambda / v$



Следствие $\oint \vec{E} d\vec{l} = 0$

$$U_2 - U_1 = \int_1^2 \vec{F}_{вект} \cdot d\vec{l}$$

$$\vec{F}_{вект} = -q \vec{E}$$

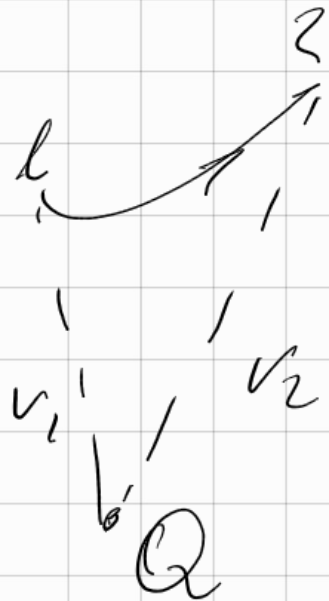
$$U_2 - U_1 = -q \int_1^2 \vec{E} d\vec{l}$$

$$\varphi = \frac{U}{q}$$

Потенциал

$$\varphi_2 - \varphi_1 = - \int_1^2 \vec{E} d\vec{l}$$

Правило суперпозиции: $\varphi = \varphi_1 + \varphi_2 + \varphi_3$



$$1) \quad Q \quad \varphi = \frac{Q}{r}$$

$$2) \quad \rho(\vec{r}) \quad dq = \rho dV$$


РАСПР. ЗАР. В ПРОСТР.

$$\varphi = \int_V \frac{\rho(\vec{r}) dV}{|\vec{R} - \vec{r}|}$$

3) Пот. гравит.

$$\varphi = \frac{\vec{P} \vec{r}}{r^3}$$

Энергия гравит. во внешнем поле:

$$U = -\vec{P} \vec{E}_{\text{внеш}} \quad \vec{P} = \text{const}$$

$$U = -\frac{1}{2} \vec{P} \vec{E}_{\text{внеш}} \quad P \propto E_{\text{внеш}}$$

$$d\varphi = -\vec{E} d\vec{l} \quad \begin{matrix} (E_x dx + E_y dy + E_z dz) \\ d\vec{l} = (dx; dy; dz) \end{matrix}$$

$$f(x) \Rightarrow df = \frac{df}{dx} dx$$

$$\frac{\partial f}{\partial x}; \frac{\partial f}{\partial y}; \frac{\partial f}{\partial z}$$

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

$$E_x = -\frac{\partial \varphi}{\partial x}$$

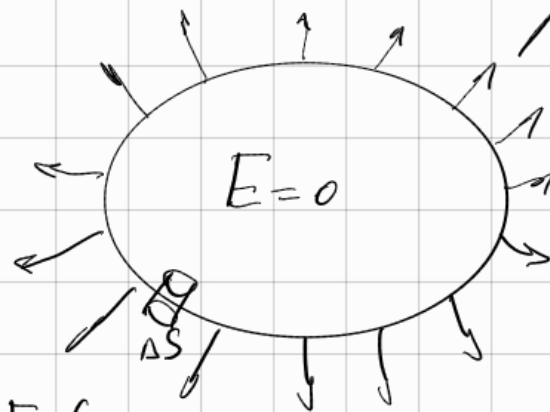
$$E_y = -\frac{\partial \varphi}{\partial y}$$

$$E_z = -\frac{\partial \varphi}{\partial z}$$

$$\vec{E} = -\text{grad } \varphi = -\vec{\nabla} \varphi$$

$$\vec{\nabla} = \left(\frac{\partial}{\partial x}; \frac{\partial}{\partial y}; \frac{\partial}{\partial z} \right)$$

Электрическое поле в проводнике равно нулю



$$j = \sigma E$$

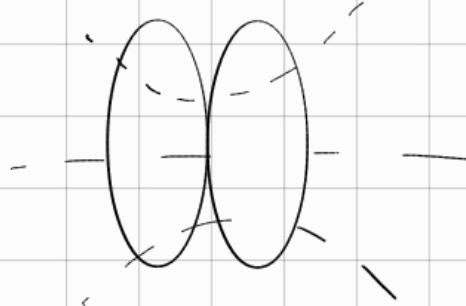
Эквипотенциальная пов-сть $\varphi = \text{const}$

$$E_{\Delta S} = n \rho \Delta S$$

$$E = n \rho \quad \text{СИ}$$

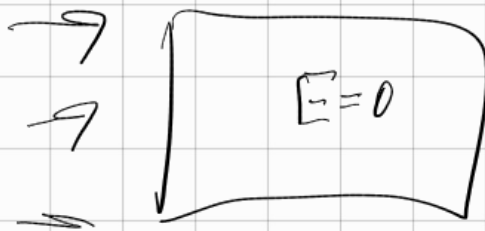
$$E = \frac{\sigma}{\epsilon_0} \quad \text{СИ}$$

$$\varphi_2 - \varphi_1 = - \int_1^2 \vec{E} d\vec{l}$$



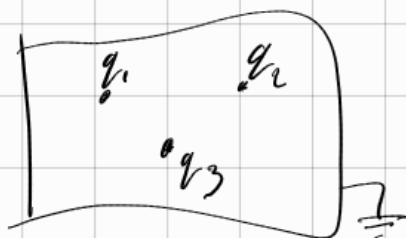
Поэтому эквипотенциальна,
а поле внутри равно нулю

Клетка Фарадея



Замкнутой цепи нет
во внешней поле
внутри поле = 0

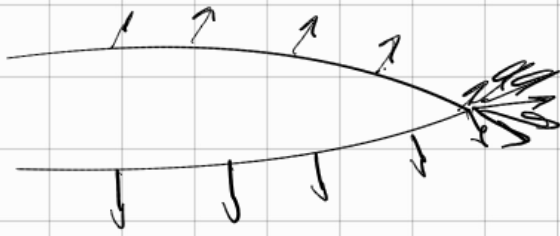
Загруженные



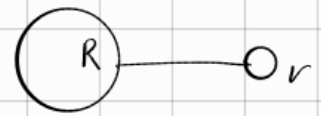
$$\varphi = 0$$

$$E = 0$$

Дл. мал. болуу олтур



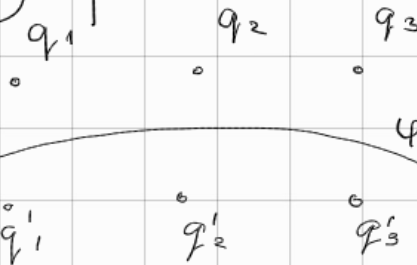
$$\varphi = \frac{Q}{R} = \frac{q}{r}$$



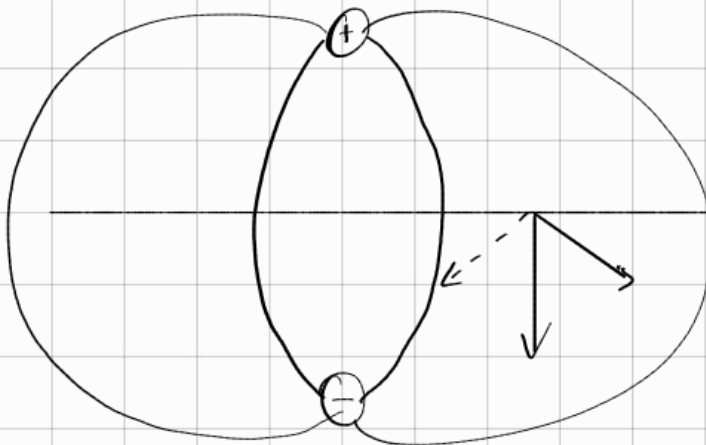
$$E = \frac{q}{r^2} \quad E_0 = \frac{Q}{R^2}$$



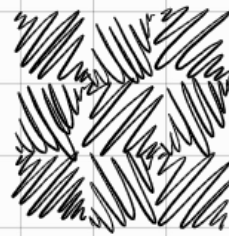
Метод изображений



08.09.26

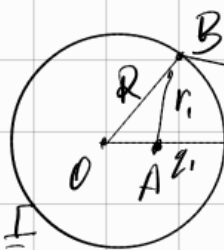


$\varphi = 0$



2. Заряд + чет. сфера

$\varphi = 0$



$OA = |a|$

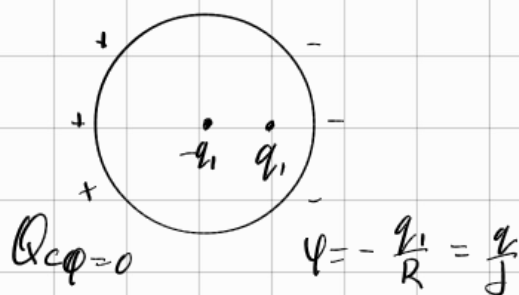
$+q$

$$Q_{сф} = q, \quad \varphi = \frac{q}{r_2} + \frac{q}{r_1} = 0 \quad q_1 = q \frac{r_2}{r_1} = -q \frac{R}{a}$$

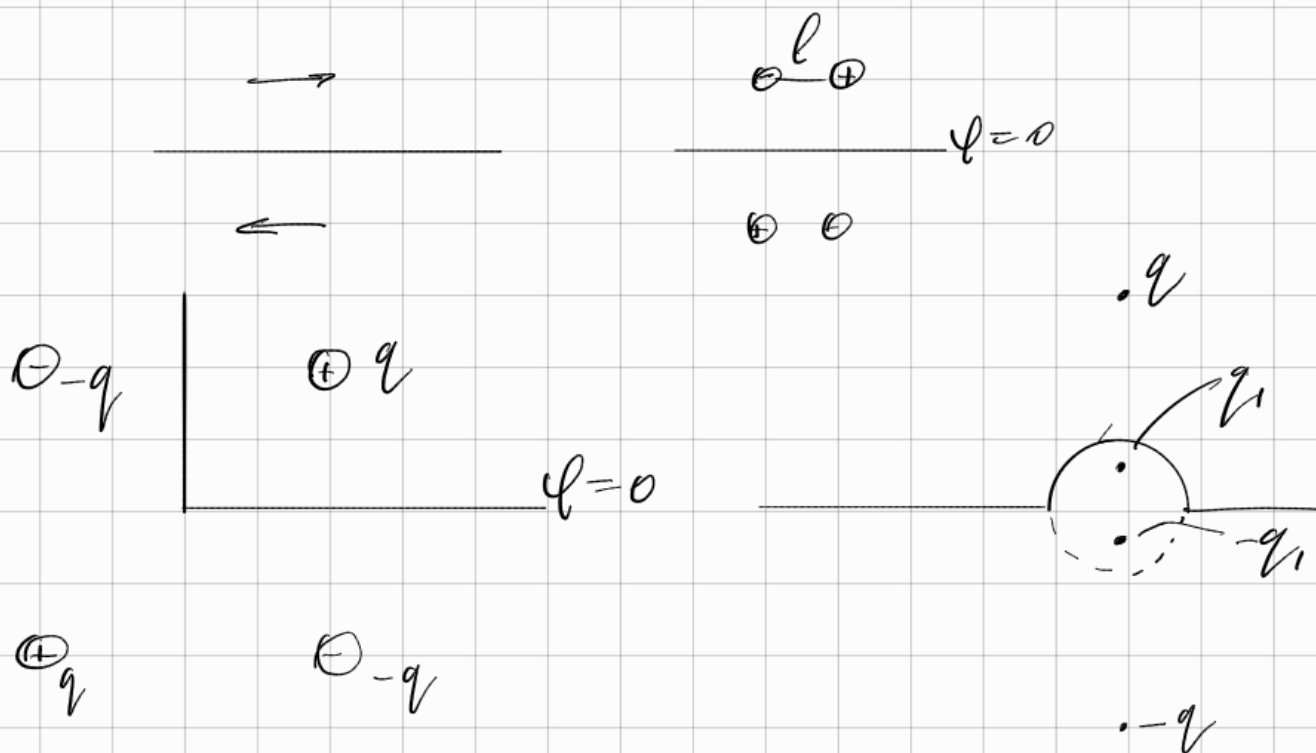
$$\Delta OBA \sim \Delta OBC$$

$$K = \frac{R}{f} = \frac{a}{R} \Rightarrow a = \frac{R^2}{f} \quad q_1 = \frac{q}{f}$$

$$F = \frac{qq_1}{(f-a)^2} \quad \text{сумма}$$



$$\frac{q}{f}$$



Метод Лаврентьева в квадрате.

$$\Phi = \oint \vec{E} \cdot d\vec{S} = 4\pi Q \quad \text{сгсз}$$

$$\vec{E} = (E_x, E_y, E_z)$$

$$d\Phi_x = E_x(x+dx, y, z) dy dz - E_x(x, y, z) dy dz$$

$$d\Phi_x = \frac{\partial E_x}{\partial x} dx dy dz = dV$$

$$d\Phi_y = \frac{\partial E_y}{\partial y} dV \quad d\Phi_z = \frac{\partial E_z}{\partial z} dV$$

$$d\Phi = d\Phi_x + d\Phi_y + d\Phi_z = 4\pi \rho dV$$

$$\vec{E} = (E_x, E_y, E_z)$$

$$d\Phi_x = E_x(x+dx, y, z) dy dz - E_x(x, y, z) dy dz$$

$$d\Phi_x = \frac{\partial E_x}{\partial x} dx dy dz = dV$$

$$d\Phi_y = \frac{\partial E_y}{\partial y} dV \quad d\Phi_z = \frac{\partial E_z}{\partial z} dV$$

$$d\Phi = d\Phi_x + d\Phi_y + d\Phi_z = 4\pi \rho dV$$

$$\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = 4\pi\rho$$

$$\vec{\nabla} \cdot \vec{E} = 4\pi\rho$$

записываем $\vec{E}$
 $\text{div } \vec{E} = 4\pi\rho$

Уравнение Пуассона

$$\vec{\nabla} \cdot \vec{E} = 4\pi\rho$$

$$\vec{E} = -\vec{\nabla}\varphi$$

$$\vec{\nabla}^2 \varphi = -4\pi\rho$$

$$\vec{\nabla}^2 = \Delta$$

$$\Delta = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\Delta \varphi = -4\pi\rho$$

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} = -4\pi\rho$$

$\rho=0$ - V-е решение

$$(\Delta \varphi = 0)$$

$\varphi = \text{const}$

Переписать у Алеова

15.09

$$\vec{\nabla} \vec{D} = 4\pi \rho_{\text{воб}} = 0$$

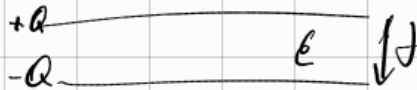
$$\vec{D} = \epsilon \vec{E}$$

$$\epsilon \nabla \vec{E} = 0$$

$$\vec{\nabla} \vec{E} = 4\pi \rho_{\text{пол}} = 0 \quad \epsilon = \text{const}$$

Энергия электростат. поля

$$U = \varphi_i - \varphi_1 \quad Q = CU$$



$$D = 4\pi \sigma$$

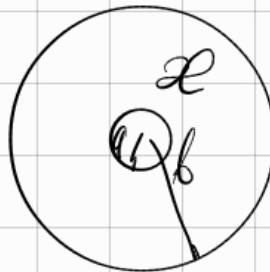
$$E = \frac{4\pi \sigma}{\epsilon} \Rightarrow \frac{U}{d} = \frac{4\pi Q}{\epsilon S} \Rightarrow$$

$$\Rightarrow Q = \frac{\epsilon S}{4\pi d} U$$

$$C = \frac{\epsilon S}{4\pi d} \quad C \propto C \propto$$

$$C = \frac{\epsilon \epsilon_0 S}{d} \quad CU$$

Узкополосный конденсатор $Q = C\varphi$
 $\varphi = \frac{Q}{R} \Rightarrow C = R$



$$D = \frac{2Q}{r}$$

$$E = \frac{2Q}{\epsilon r} \quad U = \int_a^b E dr =$$

$$= 2 \frac{Q}{\epsilon} \ln \frac{b}{a}$$

$$C = \frac{\epsilon L}{2 \ln b/a}$$

Энергия электростат. поля



$$dW = SA = U dQ = U C dU$$

$$W = \frac{CU^2}{2} \quad W = \frac{Q^2}{2C} \quad W = \frac{QU}{2}$$

① $W = \frac{1}{2} \sum_{i=1}^N Q_i \varphi_i$ — энергия электростат. поля
 $Q_i > \rho dU$

② $W = \frac{1}{2} \int \rho \varphi dU$

Термодинамическая энергия

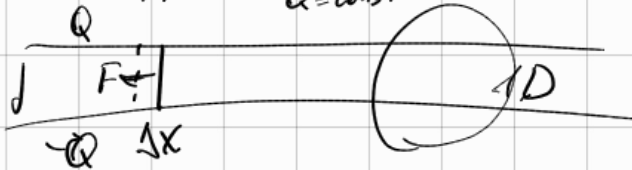
$$\vec{D} = \epsilon \vec{E}$$

$$w = \frac{\vec{E} \vec{D}}{8\pi} = \frac{\epsilon E^2}{8\pi} = \frac{D^2}{8\pi\epsilon}$$

$$W = \int_U w dU$$

Полупроводниковые конденсаторы

$Q = \text{const}$



$$F \Delta x = -\Delta W \quad W = \frac{l}{2} \frac{Q^2}{C}$$

$$F = -\left(\frac{dW}{dx}\right) Q$$

$$F = -\frac{Q^2}{2} \frac{d(1/C)}{dx} = \frac{Q^2}{2C^2} \frac{dC}{dx} = \frac{U}{2} \frac{dC}{dx}$$

Энергия. силы в конденсаторах
(вытесн. энергии)

$$\Delta W = \frac{U \Delta Q}{2}$$

$$F \Delta x = -\Delta W + \Delta A = \frac{1}{2} U \Delta Q = \Delta W$$

$$\Delta A = U \Delta Q$$

$$F = \left(\frac{dW}{dx}\right) U \quad W = \frac{CU^2}{2}$$

$$F = \frac{U^2}{2} \frac{dC}{dx}$$

$$\Delta W = \Delta U \frac{\epsilon E^2}{8\pi} - \Delta U \frac{E^2}{8\pi} = \frac{E^2}{8\pi} (\epsilon - 1) a d \cdot \Delta x$$

$$F = \frac{\epsilon E^2}{8\pi} - \frac{E^2}{8\pi} a d = (w_2 - w_1) a d \quad S_T$$

$$F = \left(\frac{dW}{dx}\right) U$$

$$P = \frac{F}{S_T} = w_2 - w_1$$

$$P = w_2 - w_1$$

C_{vac}

конд.

конд. ϵ

